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Unemployment Clubs in Russian Regions

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ABSTRACT: In this article, we empirically investigate regional unemployment in Russia. We first detect the existence of two unemployment clubs, that is, regions with high (low) unemployment surrounded by regions with high (low) unemployment, and a group that comprises the remaining regions. We then apply a specially designed class of spatial-econometric models to regional data 2005–2012, using difference GMM, and we obtain partial confirmation of our two main hypotheses: (i) spatial effects for the high-high and low-low clubs regions differ significantly; and (ii) the determinants of unemployment of the two clubs significantly differ with respect to those of the remaining regions. Our results have key implications for the national- and regional-level policies.

KEY WORDS: regional clubs, regional unemployment, Russia, spatial effects

JEL CLASSIFICATION: J64, R23, R19

Better knowledge of regional differences may enable the design of more appropriate economic and structural (national and regional) policies to alleviate the adverse socioeconomic effects associated with spatial concentrations of high unemployment (Elhorst 2003). In order to design more efficient economic policies, the heterogeneity of unemployment level has to be taken into consideration, as well as potential heterogeneity of regional unemployment factors. Hence, knowing how regions are distributed into high or low unemployment groups/clusters is a key empirical issue with significant policy implications.[1] The socio-economic development plan for the Russian Federation until 2020 states that the priorities of the state regional policy will be (i) balanced socio-economic regional development and (ii) the reduction of interregional disparities.

In the research reported by this article, we adopted a "clubs approach" to examine the degree of the homogeneity of Russian regions according to the level of regional unemployment. First of all, the exploration of Moran's scatterplots revealed a significant heterogeneity among Russian regions. In particular, among all Russian regions, we could distinguish the two main clubs: High-High (regions with high unemployment, surrounded by regions with high unemployment) and Low-Low (regions with low unemployment, surrounded by regions with low unemployment); however, some regions were not included in these two clubs.

Our main research hypotheses were: (1) spatial effects for the High-High and Low-Low clubs regions differ significantly, (2) the determinants of unemployment for the High-High and Low-Low clubs significantly differ from those for the other regions. The hypotheses were tested using a specially designed class of spatial-econometric models and estimated by regional data from 2005 to 2012 using

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Color versions of one or more of the figures in the article can be found online at www.tandfonline.com/mree.

difference GMM. Partial empirical confirmation was found for both hypotheses. Our results can favor the better design of national and regional policies to improve labor market performance and reduce regional unemployment disparities in Russia.

In the next section, we review some literature on regional labor market dynamics and Russian regional unemployment. In the third section, we describe the distinction of Russian regions into several clubs using Moran's plots and formulate two main research hypotheses. In the fourth section, we present our data sources, discuss the choice of explanatory variables, and formulate minor hypotheses concerning the influence of various factors on the unemployment rate. In the fifth section, we describe the methodology of econometric modeling. The sixth section sets out the results of the estimation and their interpretation. The last section contains some concluding remarks and policy implications.

Literature Review

In this section, we briefly recall the key literature on (i) regional real convergence and labor market dynamics and (ii) Russian regional development and unemployment.

Regional Real Convergence, Regional Labor Market Dynamics, and Club Approaches

In the past two decades, a significant amount of literature has investigated economic growth features and labor market performance dynamics by adopting regions as the key geographical dimension.

For the analyses of regional economic growth, several authors have considered the determinants of real convergence or divergence (Albu, Polimeni, and Iorgulescu 2010; Armstrong 1995; Barro and Sala-i-Martin 1991; Blanchard and Katz 1992; Boldrin and Canova 2001; Fischer 2015; Magrini 2004; Obstfeld et al. 1998), especially in the European context. In particular, Alexiadis (2013) provides a review of the theoretical and empirical literature on club convergence with applications to EU regions.

As for the literature on regional unemployment, three surveys have focused on the theoretical and empirical explanations (Elhorst 2003) and the peculiarities of transition countries (Ferragina and Pastore 2008; Huber 2007). Several studies have considered unemployment and labor market dynamics in European regions (Decressin and Fatás 1995; Francis 2009; Caroleo and Pastore 2010; Falk and Leoni 2010; Marelli and Signorelli 2010a; Beyer and Smets 2014; Marelli, Resmini, and Signorelli 2014; Beyer and Stemmer 2015; Mussida and Pastore 2015a; 2015b), also investigating the persistent differentials (Gray 2004) or considering the impact of the last crisis on NEET and/or youth unemployment (Bruno, Marelli, and Signorelli 2014; Marelli, Patuelli, and Signorelli 2012). Some research studies have focused on transition economies (Scarpetta and Huber 1995; Scarpetta and Worgotter 1995), while others have considered the regional unemployment in a specific Western (e.g., Lopez-Bazo, Del Barro, and Artis 2005) or Eastern (e.g., Tyrowicz and Wojcik 2010) European country.

Many studies on real regional growth follow a club approach (Baumont, Ertur, and Le Gallo 2003; Canova 2004; Quah 1997).[2] However, a cluster approach (Overman and Puga 2002) is more widespread in the literature on unemployment. More recently, several chapters in Mussida and Pastore (2105a) have adopted a cluster approach to investigate geographical labor market imbalances. The main technical difference between a club and cluster approach is that researchers who use a cluster approach usually try to unify regions with close values of explanatory variables (this requires the use of a special multidimensional distance between objects, for example, the Euclidean or Mahalonobis distance). Researchers who use the club approach usually unify regions with close values of the dependent variable. However, the terms "cluster" and "club" are sometimes used as synonyms. In studies for Europe and Russia, the idea of clubs has been commonly used for models of economic growth (Fisher and LeSage 2013; Kholodilin, Oshchepkov, and Siliverstovs 2012). However, from our point of view, the idea of the club approach is well suited to the study of regional unemployment levels in Russia.

Although our quite different approach does not permit comparison with studies applied to other countries, it should be noted that some of their findings (especially the existence of a significant spatial correlation and persistence over time) are coherent with some of our results for Russian regions. Here

we briefly recall the approach and results of some of the most recent studies on other countries. Firstly, it should be highlighted that the literature adopting a regional (subnational) dimension contrasts with the labor market literature based on the key role of national institutions (e.g., Nickell, Nunziata, and Ochel 2005) and ignoring any internal dynamics within countries.[3] Overman and Puga (2002) adopted a nonparametric approach and strongly documented that geographical unemployment clusters in Europe do not respect national boundaries. They also found a "neighbor effect" jointly with a significant degree of persistence. Kondo (2015) ta\kes a spatial statistical approach using Moran's I and Getis-Ord's Gi*(d) statistics and finds spatial persistence and interdependence (positive spatial autocorrelation) in Japanese unemployment by using a six-period municipal panel dataset. Korobilis and Gilmartin (2012) used a mixture panel data model to describe unemployment differentials between heterogeneous groups of UK regions and found the existence of two clusters, with a key role being played by the share of sectoral employment and the degree of urbanization. Halleck-Vega and Elhorst (2016) adopted an innovative methodology simultaneously accounting for serial dynamics, spatial dependence, and common factors; this approach was applied to provincial level data for the Netherlands and suggested the risk of a different approach in terms of biased inference. Filiztekin (2009) used a spatial and nonparametric technique to document the wide regional unemployment disparities in Turkey and found persistence and the existence of some clusters with a key role in human capital and aggregate demand deficiency. Wu (2003) used a panel data method based on the Barro approach and Edwards' work, and he focused on persistence and the role of sectoral employment in explaining provincial unemployment levels and dynamics in China.

Russian Regional Development and Unemployment

Most studies on Russian regions focus on economic growth and development features, while only a few of them investigate the unemployment differences and dynamics.

In particular, some research studies have investigated regional real convergence or divergence (Akhmedjonov, Lau, and Izgi 2013; Kholodilin, Oshchepkov, and Siliverstovs 2012; Lehmann and Silvagni 2013; Solanko 2008; Ledyaeva and Linden 2008), while others have focused on regional inequality (Dolinskaya 2002; Galbraith et al. 2004; Mikheeva 1999). In addition, Popov (1999) focused on the importance of reform strategies; Ahrend (2005) analyzed the role of the speed of reform and the initial conditions for regional economic performance; Desai, Freinkman, and Goldberg (2005) considered the effects of fiscal federalism; Oshchepkov (2015) focused on the role of wage differentials.

The literature on Russian regional labor market differences has especially focused on youth unemployment (e.g., Demidova, Marelli, and Signorelli 2013, 2015; Demidova and Signorelli 2012). To the best of our knowledge, the existing literature does not give answers about the number and features of regional unemployment clubs in Russia. In addition, questions about the possible differences in spatial effects and in the impact of explanatory variables on regional unemployment level remain open.

Regional Unemployment Clubs

A Moran's scatter plot is a useful tool for visualization in spatial econometrics. It shows the dependence between the observable variable (Y—unemployment rate for the current study) and the weighted sum of the indicator for the other regions (WY), where W is the weights matrix. Special weights matrices are used to account for the spatial relationships between regions in econometrics. There are several types of matrices with varying complexity. In our analysis, as a first strategy to account for the spatial dependence of the regions, a matrix of common borders was formed. It was represented by the following formula of W:

$$W_{len} = \begin{pmatrix} 0 & w_{12}^{len} & \dots & w_{1n}^{len} \\ w_{21}^{len} & 0 & \dots & w_{2n}^{len} \\ \vdots & \vdots & \ddots & \vdots \\ w_{n1}^{len} & w_{n2}^{len} & \dots & 0 \end{pmatrix},$$

where $w_{ij}^{len} = \dfrac{length\ in\ km\ of\ jo\,\text{int}\ boundaries\ between\ regions\ i\ and\ j}{total\ length\ in\ km\ of\ all\ boundaries\ of\ region\ i}$

We used information about the length of joint boundaries taken from the State real estate cadastre.[4]

The matrix is line-normalized, so that w_{ij} accounts for the weights of a region; $w_{ij} = 0$, if there is no boundary between regions i and j or if $i = j$. To continue the graph's description, its horizontal axis shows the values of Y and the vertical axis the weighted sum WY. The values of Y are usually considered as normalized (Z), which obtains due to centering and normalization procedures: $Z = \dfrac{Y - \mu}{\sigma}$, where μ is the mean value of Y and σ is the standard deviation of Y. The horizontal and the vertical axes (Z and WZ) divide the graph into four parts. Each part corresponds to one regional group: top right—High-High, top left—Low-High, bottom left—Low-Low, bottom right—High-Low.

It should be noted that such a graph may be constructed for each year of the period. Eight Moran's scatter plots were built to answer the question concerning the number of regional groups in Russia (see Appendix A). The first finding was that there are four regional unemployment groups in Russia. It was then necessary to study the membership of each group to determine whether it was stable over time. This procedure showed that only High-High and Low-Low groups were more or less stable over time. They were consequently called "clubs" and were later incorporated into the model by the dummy construction. All other regions were united into a group called High-Low. The rule of simple majority was used to define the region belonging to the club; if a region was more than five times in the High-High or the Low-Low group, it was labeled as High-High or Low-Low, respectively. The results are represented in Figure 1.

This map shows the division of Russian regions into four regional unemployment groups according to the Moran's scatterplots built for each year of the reporting period 2005–2012. Green stands for regions in the Low-Low club: these are prosperous regions with low unemployment rates surrounded by regions with low unemployment rates. This group is the most numerous (46 regions) and consists of all regions of the Central Federal district, the majority of the regions belonging to the North-West, Volga, and Ural Federal district. Red represents regions of the High-High club (13 regions), which consists of two regions in the Southern Federal district (Republic of Kalmykia and Astrakhan Oblast),

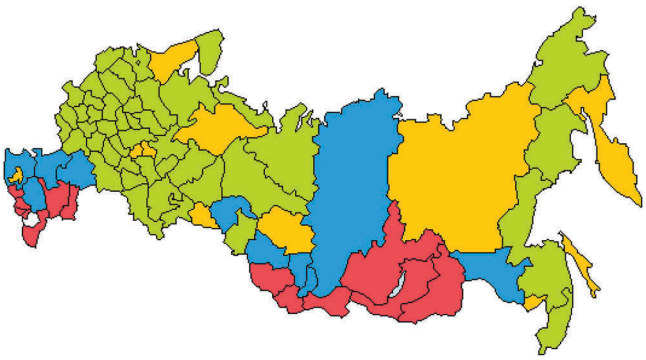


Figure 1. Regional unemployment clubs in Russia in 2005–2012.

NOTE: Regions in the High-High club are colored red; regions in the Low-Low club are colored green; regions in the Low-High group are colored blue; and regions in the High-Low group are colored yellow.

five regions in the North Caucasus Federal district (Republic of Dagestan, Republic of Ingushetia, Kabardino-Balkar Republic, Karachay-Cherkess Republic, Republic of North Ossetia-Alania), and six regions in the Siberian Federal district (Altai Republic, Republic of Buryatia, Tuva Republic, Altai Krai, Zabaykalsky Krai, Irkutsk Oblast). Regions in the Southern Federal district and in the North Caucasus district are located in the lower left part of the map, and regions from the Siberian Federal district in the lower central part. Blue regions are the regions of the Low-High group, and yellow regions belong to the High-Low one. Together, these two groups form the High-Low group.

The southern part of Russia is mostly represented by regions with high unemployment surrounded by regions with high unemployment (High-High club), while the western part of Russia turns out to be prosperous (Low-Low club). The result of this procedure defines two clubs stable over time, which are the Low-Low and High-High regional unemployment clubs.

To test the robustness of our analysis, we also used two other weights matrices: the binary contiguity matrix and the matrix of inverse distance between the capitals of the regions by auto road.

The binary contiguity matrix is a row-normalized matrix

$$W_b = \begin{pmatrix} 0 & w^b_{12} & \dots & w^b_{1n} \\ w^b_{21} & 0 & \dots & w^b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ w^b_{n1} & w^b_{n2} & \dots & 0 \end{pmatrix},$$

where $w^b_{ij} = 1$ if regions i and j have a joint boundary and 0 otherwise.

The matrix of inverse distance is a row-normalized matrix

$$W_{id} = \begin{pmatrix} 0 & w^{id}_{12} & \dots & w^{id}_{1n} \\ w^{id}_{21} & 0 & \dots & w^{id}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ w^{id}_{n1} & w^{id}_{n2} & \dots & 0 \end{pmatrix}$$

where w^{id}_{ij} is the inverse distance between the capitals of the regions by auto road (for all regions except Chukotka Autonomous Okrug ($i = 80$), Kamchatka territory ($i = 73$), and Sakhalin region ($i = 78$) because there are no traditional auto roads in these three regions). For these regions, we used modified formulas for the elements of the weights matrix, namely, $w^d_{ij} = 0.5(w^b_{ij} + w^{len}_{ij}),\ if\ i = 73,\ 78,\ 80$ and $w^d_{ij} = 0,\ if\ j = 73,\ 78,\ 80$.

The results of using these additional matrices to split regions into several clubs were similar.

In our research, we used an approach similar to the one applied by Kholodilin, Oshchepkov, and Siliverstovs (2012) to study the Russian regional convergence process. Using Moran's plot, they split all Russian regions into four parts [HH (LL)—regions with high (low) income located near similar regions, HL—high-income regions located near low-income regions, LH—low-income regions located near high-income regions] and found that speeds of convergence in the group of HH regions and LL regions differ significantly, while in the other regions, Kholodilin et al. did not detect convergence. In our previous studies (Demidova 2015; Demidova, Marelli, and Signorelli 2013), we separated Russian regions into eastern and western parts and found differences in spatial effects and determinants of youth unemployment. Here, similarly to Kholodilin, Oshchepkov, and Siliverstovs (2012) convergence analysis, we adopted a regional club approach to investigate the differences in both spatial effects and determinants of unemployment. In particular, we formulated two main research hypotheses:

1. *Spatial effects for the High-High and Low-Low clubs differ from spatial effects for other regions;*
2. *The determinants of unemployment for the High-High and Low-Low clubs differ from those for other regions.*

In the next sections, we verify these hypotheses.

Data, Dependent and Explanatory Variables

Our sample consisted of 80 regions. The majority of the data used in the research was available for public access via the website of the Federal State Statistics Service (FSSS) of the Russian Federation. The information covered the period of eight years from 2005 to 2012. It was impossible to include earlier years in the research due to the different classification of industries before the year 2005. Moreover, data on some regions were not included in the study because of its absence (the Republic of Chechnya, the Republic of Crimea, and Sevastopol). In addition, the Kaliningrad region was not included in the study because it has no common borders with other regions of Russia. Moreover, during the reporting period, some regions underwent changes of an administrative-territorial character. This altering of boundaries was taken into consideration, mitigated by an aggregating procedure (see Appendix B).

The dependent variable was the regional unemployment rate. Its dynamic for the reporting period is shown below (see Figure 2). Until the year 2007, the average unemployment level in Russia declined. Then, during the crisis years, it slightly rose. By the end of the period, it had decreased by about one third compared with the initial value, reaching the level of 6.77 percent.

The graph also shows the average unemployment rate in regional clubs. The dynamics of the average unemployment rate in regional clubs (High-High and Low-Low) and the High-Low regional group are similar to the country indicator, with a gap of 10–11 percentage points. The existence of these persisting differences was an incentive for the current study; these similar dynamic trends made separate dummy variables for each year in our econometric estimations unnecessary.

To explain existing levels of unemployment rate and to test two main research hypotheses, three groups of variables were chosen: (1) variables about the attractiveness of the region; (2) socio-demographic variables; and (3) variables of the industrial structure of the employed population.

The first group of variables consisted of four indicators: (i) GRP per capita (variable *grp*, thousand rubles in prices of basic year), (ii) productivity per worker (variable *productivity*, thousand rubles in prices of basic year), (iii) the share of urban population (variable *urban_share*, %), and (iv) population density (variable *density*, %). According to numerous studies (e.g., Marelli and Signorelli 2010b), growth in GRP per capita and, often, also productivity per worker are expected to lower the unemployment rate. As for the population density, Elhorst (2003) showed that it is difficult to make an unambiguous assumption about its effect on the unemployment are. The negative-sign effect (decrease of the unemployment rate) may be explained by the following reason: the higher the population density, the more efficient is the job search process. However, an opposite-sign effect exists: the higher the population density, the more difficult it is to find a job due to the increased competition. In addition, the effect of population density on unemployment may differ from country to country (Blackley 1989; Partridge and Rickman 1995; Elhorst 2003).

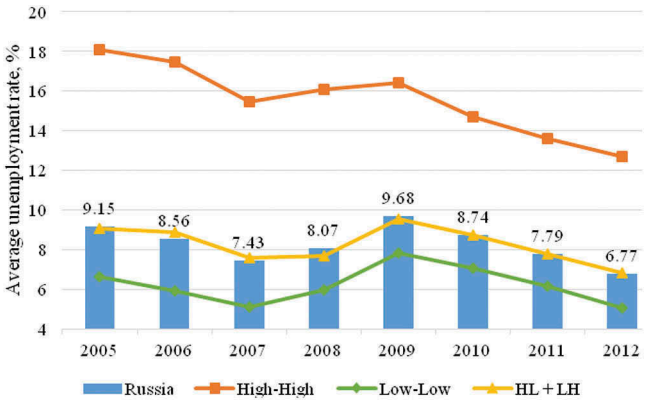


Figure 2. The dynamic of average unemployment rate in Russia for 2005–2012, %.

The socio-demographic features of the population consist of variables characterizing the age structure of the population and the stock of human capital. To illustrate the age structure, the shares of people below and above working age were taken as variables (*below_working* and *above_working*, %). Working age in Russia is above 16 and below retirement age, which is 60 years for men and 55 for women. The age structure of the population is expected to affect the unemployment rate in the following way: a high proportion of young people leads to an increase in unemployment (Elhorst 1995), as well as to a high proportion of older people.

The stock of human capital was measured as the share of the employed population with a higher education (variable high_education, %), where "higher education" means that someone has gained at least a higher professional education according to the FSSS classification of educational levels. A high proportion of people with higher educations is expected to lower unemployment rates. This result has been empirically confirmed by several studies (Burridge and Gordon 1981; Siegers 1983; Simon 1988; Holzer 1993; Malizia and Ke 1993; Partridge and Rickman 1995). Possible explanations for this trend are presented in Elhorst (2003) and Semerikova (2015): educated people have more job offers, and they are better informed about the situation in the labor market. Both speed up the job search process, which decreases unemployment.

According to Martin (1997), the sectoral structure is detected as one of the most important groups of variables that explain the unemployment at the regional level. Its importance may be accounted for as follows (see also Elhorst 2003; Partridge and Rickman 1995): i) the same economic sector can have different levels of unemployment in different regions; moreover, one or more sectors may be characterized by a generally higher or lower level of regional unemployment compared with others.

In our research, the sectoral structure of the employed population consisted of six separate variables: the share of employment in agriculture (variable *agriculture*, %), the share of employment in construction (variable *construction*, %), the share of employment in wholesale and retail trade (variable *trade*, %), the share of employment in the public sector (consisting of education and health services, variable *public*, %), the share of employment in mining (variable *mining*, %), and the share of employment in manufacturing (variable *manufacturing*, %). As the reference category, we used the share of employment in services.

Based on the expected direction of these factors' influence on regional unemployment rates, several minor hypotheses were put forward:

3. *The higher the GRP per capita (or productivity per worker), the lower the unemployment rate*;
4. *The higher the share of urban population, the lower the unemployment rate*;
5. *The higher the share of people below working age, the higher the level of unemployment*;
6. *The higher the share of people above working age, the higher the level of unemployment*;
7. *The higher the share of educated population, the lower the unemployment rate*.

Because population density may influence the unemployment rate in two directions, there was no minor hypothesis for this factor. Moreover, there were no expectations (or minor hypotheses) about the influence of the sectoral structure.

Table 1 above shows the descriptive statistics of the explanatory variables. There are disparities between High-High and Low-Low clubs. For example, all variables that characterize the attractiveness of the region are higher in the Low-Low club. Moreover, the share of the agricultural sector is notably higher in regions belonging to the High-High club, while the share of the mining and manufacturing sector is considerably higher in the regions of the Low-Low club. The table illustrates the inequality among Russian regions, which is consistent with the earlier observation based on the data on the unemployment rate (see Figure 2).

To test whether we should take account of the spatial heterogeneity of Russian regions, we calculated Moran's indexes for the three weights matrices that we mentioned above: the binary contiguity one, the matrix of boundary lengths, and the matrix of inverse distance between the capitals of the regions by road (see Table 2).

The Moran's indexes were significant for all years (except 2005) for the binary contiguity weights matrix and, for most years, for the boundary length matrix and matrix of inverse distance. It is for this reason that it was necessary to include spatial lags in our model. Note that Moran's indexes decreased

Table 1. Descriptive statistics of explanatory variables in regional clubs.

	Variable	productivity	GRP	density	urban_share	below_working	above_working	high_education	agriculture	construction	trade	public	mining	manufacturig
All regions	Mean	254.7	129.5	71.4	69.1	17.2	20.5	23.6	12.2	7.2	15.6	17.0	2.4	14.8
	Std. Dev.	188.0	134.5	370.2	12.6	3.4	4.7	7.0	5.6	2.4	3.5	3.1	4.3	6.4
	Min	87.6	14.8	0.1	26.0	12.1	5.9	1.6	0.4	3.5	4.0	10.9	0.0	1.0
	Max	1694.3	1282.8	3538.0	100.0	32.2	28.6	46.0	28.1	20.8	24.1	34.5	28.8	30.9
High—High Club	Mean	166.9	65.3	33.0	53.4	22.1	16.0	26.2	17.0	6.4	13.7	21.1	1.7	10.2
	Std. Dev.	41.7	23.8	40.4	13.2	4.4	4.1	7.1	5.9	1.7	2.1	4.6	2.0	4.8
	Min	87.6	14.8	1.8	26.0	15.8	8.0	6.0	7.4	3.5	9.0	16.3	0.2	2.9
	Max	308.6	144.9	143.5	79.6	32.2	24.1	46.0	28.1	11.8	21.1	34.5	11.3	17.9
Low—Low club	Mean	283.5	152.1	105.6	74.0	15.7	22.0	23.4	10.9	7.5	15.9	15.7	2.5	17.1
	Std. Dev.	229.6	167.1	484.9	9.8	2.2	4.6	7.0	5.5	2.9	3.8	1.6	5.3	6.5
	Min	94.4	41.0	0.1	57.4	12.1	5.9	1.6	0.4	3.5	4.0	10.9	0.0	1.0
	Max	1694.3	1282.8	3538.0	100.0	23.4	28.6	45.8	26.0	20.8	24.1	19.4	28.8	30.9

Table 2. Dynamics of Moran's spatial correlation index for the variable *Unemployment*.

Weights matrix	Boundary lengths matrix	Binary contiguity matrix	Inverse distance matrix
2005	0.059	0.072	0.055*
2006	0.088	0.120**	0.112*
2007	0.163***	0.191***	0.158***
2008	0.134**	0.145***	0.121***
2009	0.098*	0.101**	0.088*
2010	0.09	0.098**	0.069
2011	0.081	0.085*	0.063
2012	0.110**	0.118**	0.071

Note: *Significant at 10%, **significant at 5%, ***significant at 1%.
Source: Our elaboration on Rosstat data.

for the period of 2009–2011. We can assume that the mutual influence of the regions weakened due to the global economic crisis. However, in 2012, the Moran's indexes started to rise again, indicating the recovery of links between regions.

There are many types of spatial models (Elhorst 2014), and we discuss the choice of the ones most suitable for our research in the next section.

Methodology of Econometric Modeling

We start with the most general dynamic spatial panel data model:

$$Y_{it} = \tau Y_{it-1} + \rho W_Y Y_{it} + X\beta_{it} + W_X X\theta it + \alpha_i + c_t + \varepsilon_{it}, \quad \varepsilon_{it} = \lambda W_\varepsilon \varepsilon_{it} + u_{it} \tag{1}$$

where $i = 1, \ldots, 80$- number of the region, $t = 2005, \ldots, 2012$- year, Y is the dependent variable (unemployment rate in our research), X is the matrix of explanatory variables, W_Y is the weights matrix for the autoregressive component, W_X is the weights matrix for the spatially lagged explanatory variables, W_ε is the weights matrix for the idiosyncratic error component, α_i is the individual fixed or random effect and c_t is the time effect, u_{it} is a normally distributed error term, τ is a time correlation coefficient, ρ is a spatial autocorrelation coefficient, β and θ are vectors of estimated coefficients.

We estimated several special cases of this model and compared the Akaike information criterion (AIC) and the Bayes information criterion (BIC) for all estimated models (see the details and the results of the estimation in Table 3). In all the models, we used $W_X = W_Y = W_\varepsilon$. As W we consequently used the matrix of common lengths, the binary contiguity weights matrix, and the matrix of inverse distance. Both the information criteria for all matrices were minimal for the spatial Durbin model with fixed effects. Therefore, we chose this model as the benchmark for our research.

To test the two main research hypotheses and five minor hypotheses, we used the following modification of the SDM model:

$$\begin{aligned} Y_{it} = \tau Y_{it-1} + \rho_H (W \cdot H) Y_{it} + \rho_L (W \cdot L) Y_{it} + \rho_{HL} (W \cdot (I - H - L)) Y_{it} + X\beta_{it} + \\ + (L \cdot X)\theta_{Lit} + (H \cdot X)\theta_{Hit} + \alpha_i + c_t + u_{it} \end{aligned} \tag{2}$$

where H and L are diagonal matrices characterizing membership in the High-High or Low-Low class,

Table 3. Information criteria for special models.

Type of model	Restrictions in model (1)	Type of weights matrix					
		Wleng		Wbound		Winvdist	
		AIC	BIC	AIC	BIC	AIC	BIC
SAR dynamic model with FE	$\theta = \lambda = 0$	1801	1892.17	1794.426	1885.313	1791.205	1882.09
SAR static model with FE	$\theta = \lambda = \tau = 0$	2237.94	2327.16	2234.98	2324.21	2225.78	2315.01
SDM dynamic model with FE	$\lambda = 0$	**1799.21**	**1942.04**	**1783.07**	**1925.89**	**1785.94**	**1928.769**
SDM static model with FE	$\lambda = \tau = 0$, α_i are FE	2221.62	2364.38	2206.93	2349.70	2188.42	2331.19
SDM static model with RE	$\lambda = \tau = 0$, α_i are RE	2668	2819.82	2657.01	2808.7	2667.06	2818.75
SAC static model with FE	$\theta = \tau = 0$	2182.17	2275.86	2184.486	2278.177	2206.70	2300.4
SEM static model with RE	$\rho = \theta = \tau = 0$	2678.6	2776.75	2674.602	2772.754	2664.47	2762.62

$$H_{ii} = \begin{cases} 1, & if \quad i \in \{29, 31, 34-38, 60-62, 64, 65, 67\} - \\ numbers\ of\ regions\ from\ High - High\ C\ \text{lub} \\ 0, otherwise \end{cases}$$

$$L_{ii} = \begin{cases} 1, & if \quad i \in \{1-17,\ 20-27,\ 40,\ 42-44,\ 46-53,\ 55,\ 57-59,\ 70,\ 74-75,\ 77,\ 80\} - \\ numbers\ of\ regions\ from\ Low - Low\ C\ \text{lub}, \\ 0, otherwise \end{cases},$$

I is a unit matrix,

Y is the unemployment rate, $X = (X_1 \quad X_2 \ldots X_{12})$ is the matrix of explanatory variables, $X_1 = productivity = (productivity_1, \ldots, productivity_{80})$ (or $X_1 = gdp = (gdp_1, \ldots, gdp_{80})$), $X_2 = density = (density_1, \ldots, density_{80})$,..., $X_{12} = manufacturing = (manufacturing_1, \ldots, manufacturing_{80})$, ρ_H, ρ_L, ρ_{HL} are the coefficients of the spatial lags for the High-High and the Low-Low clubs and for the High-Low group, respectively; $\beta = (\beta_{productivity}, \beta_{density}, \ldots, \beta_{manufacturing})^T$, $\theta_L = (\theta_{Lproductivity}, \theta_{Ldensity}, \ldots, \theta_{Lmanufacturing})^T$, $\theta_H = (\theta_{Hproductivity}, \theta_{Hdensity}, \ldots, \theta_{Hmanufacturing})^T$ are the vectors of coefficients.

In this model, the spatial lag of the dependent variable is split into three parts, and we use two specific spatial lags for the explanatory variables.

This form is very convenient for testing our hypothesis. Let us demonstrate this. In fact, equation (2) may be rewritten as follows:

$$\begin{pmatrix} Y_{ih} \\ Y_{il} \\ Y_{ilh} \end{pmatrix}_t = \tau \begin{pmatrix} Y_{ih} \\ Y_{il} \\ Y_{ilh} \end{pmatrix}_{t-1} + \rho_H \begin{pmatrix} WY_{ih} \\ 0 \\ 0 \end{pmatrix}_t + \rho_L \begin{pmatrix} 0 \\ WY_{il} \\ 0 \end{pmatrix}_t + \rho_{HL} \begin{pmatrix} 0 \\ 0 \\ WY_{ihl} \end{pmatrix}_t + \\ + \begin{pmatrix} X_{ih} \\ X_{il} \\ X_{il} \end{pmatrix}_t \beta + \begin{pmatrix} X_{ih} \\ 0 \\ 0 \end{pmatrix}_t \theta_H + \begin{pmatrix} 0 \\ X_{il} \\ 0 \end{pmatrix}_t \theta_L + \begin{pmatrix} \alpha_{ih} \\ \alpha_{il} \\ \alpha_{ihl} \end{pmatrix} + c_t + \begin{pmatrix} u_{ih} \\ u_{il} \\ u_{ihl} \end{pmatrix}_t \tag{3}$$

where $i_h \in \{29, 31, 34–38, 60–62, 64–65, 67\}$; $i_l \in \{1–17, 20–27, 40, 42–44, 46–53, 55, 57–59, 70, 74–75, 77, 80\}$; $i_{hl} \in \{18–19, 28, 30–31, 33, 39, 41, 45, 54, 56, 63, 66, 68–69, 71–73, 76, 78–79\}$, which are the numbers of the regions in the High-High, Low-Low, High-Low clubs (the list of all regions with numbers is given in Appendix C).

As said, two main research hypotheses may be formally tested (see the details below).

Hypothesis 1. There are no differences of spatial effects in regional clubs.

Alternative hypothesis 1. There are differences of spatial effects in regional clubs.

Formal main and alternative hypotheses 1:

$$H_0 : \rho_h = \rho_l = \rho_{hl}$$

$$H_1 : \rho_h \neq \rho_l \, or \, \rho_h \neq \rho_{hl}.$$

Hypothesis 2. There are no differences in the influence of the factors on unemployment rates in the regions belonging to different regional clubs.

Alternative hypothesis 2. There are differences in the influence of the factors on unemployment rates in the regions belonging to different regional clubs.

Formal main and alternative hypotheses 2:

$$H_0 : \theta_H = \theta_L = 0$$

$$H_1 : \theta_H \neq 0 \, or \, \theta_L \neq 0.$$

The split spatial lags in our model were endogenous. To solve the problem of endogeneity difference, GMM (Arellano and Bond 1991) is usually used as the method of estimation.

However, application of this method to the initial specification (with all explanatory variables, divided into several parts) required a huge number of instruments (more than the number of regions). According to Roodman (2009), this leads to a bias in the parameter estimation. To avoid this problem, we had to use the Arellano-Bond approach for the estimation and strictly restrict the number of instruments.

Moreover, a problem arises relative to the bias in the parameters' estimation at a small time interval but with a large number of observation units. However, Lee and Yu (2010) suggested a GMM modification for models with fixed effects that is able to solve the problem of the estimation bias. Consequently, we use this specification in our study.

A large number of variables may also lead to the problem of multicollinearity of the data. The multicollinearity consequences are the insignificance of the coefficients of many factors (which complicates interpreting the corresponding factors' effect on the dependent variable) as well as instability of estimations. For this reason, we did not include the GRP per capita and productivity variables in the same model simultaneously. To increase the efficiency of the estimates, we removed groups of insignificant variables from the model one by one (after a preliminary test of the corresponding statistical hypotheses). The technique that we used was an extension of the conventional backward stepwise method.

In detail, as the first step, we estimated model (3).

As the second step, we tested the hypothesis

$$H_0 : \theta_{Hproductivity} = \theta_{Lproductivity} = 0. \tag{4}$$

Acceptance of hypothesis (4) corresponded to the absence of differences in the influence of productivity on the unemployment rate for High-High, Low-Low, and High-Low clubs. If hypothesis (4) was rejected, we tested the hypothesis

$$H_0 : \theta_{Hproductivity} = 0. \tag{5}$$

Acceptance of hypothesis (5) corresponded to the absence of differences in the influence of productivity on the unemployment rate for High-High and High-Low clubs.

If hypothesis (5) was rejected, we tested the hypothesis

$$H_0 : \theta_{Lproductivity} = 0. \tag{6}$$

Acceptance of hypothesis (6) corresponded to the absence of differences in the influence of productivity on the unemployment rate for Low-Low and High-Low clubs.

As the third step, we estimated model (3) (if all hypotheses (4)–(6) were rejected) or model (3) without variables HX_1 and LX_1 (if hypothesis 4 was not rejected) or model (3) without variable HX_1 (if hypothesis 5 was not rejected) or model (3) without variable LX_1 (if hypothesis 6 was not rejected). After that, we repeated step (2) for the variables $X_2 = density, \ldots, X_{12} = manifacturing$ manufacturing. Finally, we tested the hypothesis $H_0 : \rho_h = \rho_l = \rho_{hl}$.

We discuss the results of the estimation in the next section.

The Results of the Estimation

The final results of the estimation using the Arellano-Bond GMM are presented in Table 4.

For each model, we also present the results of post-estimation procedures. Estimates of the coefficients obtained by the Arellano-Bond method are consistent under the following conditions (Greene 2012, p. 400): (1) errors u_{it} must be serially uncorrelated; (2) moment conditions (consisting in the orthogonality of the errors and instruments) must be correct. We remind the reader that when the Arellano-Bond approach is used, equations in difference are estimated. This makes it possible to avoid the endogeneity problem because individual effects are eliminated. It is for this reason that the errors in the difference equation must be identified as first-order autocorrelations and not revealed as higher-order autocorrelations (the corresponding AB tests are named in honor of Arellano and Bond). The second condition is verified by the Sargan test of instruments' validity. We verified these two conditions for all the models estimated.

In spatial econometric models, it is necessary to distinguish between direct and indirect effects. Direct effects indicate the effect of the explanatory variable from regions of some particular club on the unemployment rates in regions of the same club. The indirect effects show the influence of the explanatory variable from regions of one club on the unemployment rates in regions of the other club. The calculation of these effects is quite complicated and usually provides for the interpretation of the results over the long term. In our research, we investigated the short-term interpretation, so that explanation of the signs behind the explanatory variables (β) is enough because they comply with the signs of the direct effects. According to the results of autocorrelation AB and Sargan tests, the conditions for the estimate consistency are satisfied.

The first hypothesis about differences in spatial effects for regions belonging to different unemployment clubs received partial empirical confirmation. A positive spatial effect for the Low-Low and High-Low clubs was found for all spatial matrices. A spatial effect for the High-High club was significant only for the inverse distance matrix. Thus, if in some region of the Low-Low or High-Low club the economic situation improves and, as a result, the unemployment rate decreases, the situations in neighboring regions also improve because of decreased unemployment rates. A possible partial explanation for the High-High club is that the situation in the Moscow labor market influences the situation in all other regions including regions belonging to the High-High club. However, this hypothesis requires further investigation.

The second hypothesis also received partial empirical confirmation. We found club effects for the variables *productivity, grp, density, agriculture, construction*, and *public*. The more detailed

Table 4. The results of estimations for the modified SDM (spatial Durbin model).

Variable	Model1	Model2	Model3	Model4	Model5	Model6
Y_t = unemployment						
Time lag = Y_{t-1}	0.350***	0.349***	0.361***	0.380***	0.393***	0.402***
Spatial lags						
$W_{len}HY$	–0.068			–0.123		
$W_{len}LY$	0.392***			0.385***		
$W_{len}(I-H-L)Y$	0.501***			0.506***		
W_bHY		–0.016			–0.073	
W_bLY		0.524***			0.520***	
$W_b(I-H-L)Y$		0.721***			0.723***	
$W_{invdist}HY$			0.290**			0.278**
$W_{invdist}LY$			0.641***			0.620***
$W_{invdist}(I-H-L)Y$			0.707***			0.677***
productivity_hh	–0.038***	–0.039***	–0.028***			
productivity	0.002	0.002	–0.001			
productivity+productivity_hh (see Note A below)	*–0.036****	*–0.037****	*–0.029****			
grp_hh				–0.102***	–0.103***	–0.071***
grp				0.005**	0.004**	0
grp+grp_hh				–0.097***	–0.099***	–0.071***
density_hh	–0.534**	–0.636**	–0.600**	–0.492*	–0.596**	–0.568*
density_ll	–0.494*	–0.592**	–0.577**	–0.492*	–0.590**	–0.567*
density	0.497*	0.595**	0.579**	0.494*	0.592**	0.569*
density+density_hh	*–0.037***	*–0.041***	*–0.021*	*0.002*	*–0.004*	*0.001*
density+density_ll	*0.003**	*0.003***	*0.002*	*0.002***	*0.002***	*0.002*
urban_share	0.061	0.078	0.064	0.08	0.088	0.06
below_working	0.460*	0.544**	0.963***	0.460*	0.584**	0.978***
above_working	0.072	–0.018	–0.237	0.091	–0.004	–0.262
high_education	0.006	0.007	0.020*	0.004	0.006	0.021*
agriculture_hh	–0.491**	–0.416*	–0.204	–0.487**	–0.357	–0.135
agriculture	0.099	0.081	0.068	0.095	0.076	0.065
agriculture+agriculture_hh	*–0.392**	*–0.335*	*–0.136*	*–0.392*	*–0.281*	*–0.07*
construction_hh	0.878***	0.816***	0.763***	1.002***	0.962***	0.894***
construction	–0.053	0.007	0.133	–0.068	–0.008	0.106
bild+bild_hh	*0.825****	*0.823****	*0.896****	*0.934****	*0.954****	*1****
trade	0.156**	0.112	0.134*	0.151**	0.11	0.142*
public_hh	1.372***	1.173***	1.453***	1.398***	1.206***	1.404***
public_ll	0.790**	0.513	0.266	0.797**	0.511	0.277
public	–0.153	–0.014	0.177	–0.171	–0.008	0.187
public+public_hh	*1.219****	*1.159****	*1.63****	*1.227****	*1.198****	*1.591****
public+public_ll	*0.637****	*0.499***	*0.443**	*0.626****	*0.503***	*0.464**
mining	–0.092	–0.127*	–0.182**	–0.086	–0.127	–0.189**
manufacturig	–0.095**	–0.101**	–0.133***	–0.094**	–0.102**	–0.134***
d2007	–0.412*	–0.163	–0.054	–0.420*	–0.161	–0.096
d2008	0.254	0.414	1.671***	0.263	0.461*	1.647***
d2009	0.964***	0.883**	2.656***	0.974***	0.891**	2.668***
d2010	–0.349	–0.279	–0.083	–0.429	–0.39	–0.126
d2011	–0.661	–0.441	–0.068	–0.739	–0.535	–0.146
d2012	–1.084	–0.696	0.407	–1.136	–0.766	0.314
_cons	–23.04**	–23.77***	–28.63***	–24.56***	–25.95***	–28.59***
Number of instruments	50	50	50	50	50	50

(*Continued*)

Table 4. The results of estimations for the modified SDM (spatial Durbin model). (Continued)

Variable	Model1	Model2	Model3	Model4	Model5	Model6
p-v AB(1) test	0	0	0	0	0	0
p-v AB(2) test	0.91	0.688	0.747	0.914	0.863	0.582
p-v AB(3) test	0.991	0.957	0.623	0.634	0.615	0.347
p-v Sargan test	0.38	0.417	0.132	0.356	0.383	0.153
Tested hypothesis (see Note B below)	p-v	p-v	p-v	p-v	p-v	p-v
$\beta_{productivity} + \theta_{Hproductivity} = 0$	0	0	0			
$\beta_{grp} + \theta_{Hgrp} = 0$				0	0	0
$\beta_{density} + \theta_{Hdensity} = 0$	0.05	0.037	0.161	0.89	0.85	0.96
$\beta_{density} + \theta_{Ldensity} = 0$	0.065	0.032	0.131	0.046	0.018	0.13
$\beta_{agriculture} + \theta_{Hagriculture} = 0$	0.09	0.15	0.457	0.1	0.24	0.72
$\beta_{construction} + \theta_{Hconstruction} = 0$	0	0	0	0	0	0
$\beta_{public} + \theta_{Hpublic} = 0$	0	0	0	0	0.002	0
$\beta_{public} + \theta_{Lpublic} = 0$	0	0.036	0.069	0.004	0.034	0.063

Note : The sums of variables in italics (e.g. *productivity* + *productivity_hh*) were not included in our models. We have calculated the sums of the corresponding coefficients and inserted them in Table 4 to assist interpretation of the results. We have checked these hypotheses for testing the significance of the corresponding sums of coefficients; see reference 3. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

explanations are given below in terms of the minor hypotheses (3–7) and in relation to factors whose expected influence was not defined (population density and industrial structure).

We obtained empirical confirmation of our third hypothesis (*the higher the GRP per capita or productivity per worker, the lower the unemployment rate)* only for the High-High club. This result is consistent with those of some studies (e.g., Marelli and Signorelli 2010b) detecting a "virtuous model of growth" (with both production/productivity and employment increases) in the best performing economies, while other models ("extensive or intensive") may prevail in other cases.

Hypothesis 4 did not receive empirical confirmation: the coefficient of the variable *urban_share* (the share of urban population) was insignificant. As we expected, we did not obtain stable results concerning the influence of population density on the unemployment rate. The coefficient of *density* was positive for the High-Low club (in all models), positive (but less in absolute value) for the Low-Low club, and insignificant for the High-High club (in most of the models). Therefore, the effect of population density on unemployment may differ not only from country to country but also among the regions of the same country. Note that the insignificance of the coefficients for the variables *urbanization* and *population density* has also been empirically demonstrated in (Bernal-Verdugo, Furceri, and Guillaume 2012) on a panel of data for 97 countries from 1985 to 2008.

Hypothesis 5 received partial empirical confirmation. The increasing share of young people raises unemployment in the regions: as expected, this factor did not demonstrate a club effect. At the same time, the share of the elderly does not affect the level of unemployment (contradicting our hypothesis 6). The extremely complex role of population age composition has been partly considered by Hofler and Murphy (1989), Partridge and Rickman (1995), and, more recently, by Perugini and Signorelli (2007) in a regional perspective, and Choudhry, Marelli, and Signorelli (2016) in a national perspective focused on the effects on productivity.

Hypothesis 7 (the higher the share of educated population, the lower the unemployment rate) also did not receive empirical confirmation: the coefficient of the corresponding variable was insignificant. Nor did we find a club effect for this factor.[5] This coincides with the results obtained by Blackley (1989) for the US data 1970–1986.

Variables characterizing sectoral structure exhibited a stronger club effect.

We found negative coefficients for the share of people employed in agriculture for the High-High club and positive coefficients of this variable for other clubs; but these coefficients were insignificant in most of the models. For other countries, researchers have reported both positive and negative effects (for details, see Elhorst 2003).

The increased share of employment in the construction industry raises unemployment only in the High-High club. This may be partly explained by the fact that the construction industry usually attracts low-skilled workers, often migrants from CIS countries, with a high incidence of "irregular employment." Moreover, the construction sector has been the one most negatively affected by the 2008 global financial crisis and the following Great Recession.

The coefficient of the variable *trade* (i.e., private services) was positive in most of the models and did not exhibit a club effect. The interpretation of this result is similar to the previous one for construction. Note that Hofler and Murphy (1989) obtained the opposite result for the United States.

The increased share of employed people in the public sector, which included the education and health sectors, increases unemployment rates in both the High-High and Low-Low clubs, but more so in the former. This may be partly explained by a higher reservation wage and competition among high-skilled workers looking for jobs in this sector. Note that a similar result (an increase in the unemployment rate with an increase in the share of people employed in the public sector) has been obtained by many empirical studies (e.g., Hofler and Murphy 1989; Partridge and Rickman 1995; Taylor and Bradley 1997).

The coefficient of the variable *mining* was negative (but insignificant in half of the models) and did not exhibit a club effect. At the same time, the coefficients of the variable *manufacturing* were highly significant and negative in all models (but nor did this factor exhibit a club effect). Hence, the regional unemployment rate can be reduced by increasing the share of employment in manufacturing. Elhorst (1995) obtained a similar result for Europe; for the United States, several authors have reported the opposite result (Summers, Abraham, and Wachter 1986; Hofler and Murphy 1989; Partridge and Rickman 1995).

Conclusions

In this article, we have investigated the spatial effects for the regional unemployment clubs in Russia and the differences in the impact factors that explain regional unemployment. The panel data collected covered the period 2005–2012. To account for potential spatial effects, the spatial lag of the dependent variable (WY) was included in the model. All regions were divided into three groups—two clubs, High-High and Low-Low, and the group of other regions. In addition, in order to separate regions into clubs, Moran's scatter plots were constructed for each year of the chosen period. Clubs were groups of regions with low (high) unemployment surrounded by regions with low (high) unemployment, called Low-Low (High-High) club. The modifications of the Durbin spatial model (DSM) with fixed effects were estimated using difference GMM.

The main conclusions obtained are the following:

1. There are four regional groups in Russia, but only two of them are stable over time—High-High and Low-Low. For this reason, they were included in the model as clubs, while the remaining regions were grouped as High-Low.
2. Model evaluation partially confirmed the first hypothesis. So far, a positive spatial effect was detected for regions in the Low-Low club and High-Low group for all weights matrices. A spatial effect for the High-High club was significant only for the inverse distance matrix (reflecting the links between all regions).
3. The second hypothesis was also partially confirmed. We found that the determinants of unemployment for the High-High and Low-Low clubs significantly differ from those for the other regions.
4. Among all factors that influence unemployment, a group of factors that significantly increase unemployment may be defined. This group consists of the share of young people (in the whole country, i.e., for all the clubs/groups), the share of people employed in the construction sector

(in the High-High club), and the share of people employed in the public sector (in both High-High and Low-Low clubs but with different degrees of influence).

We can also distinguish a group of factors that help to reduce unemployment. It consists of growth of productivity per worker and GRP per capita (in the High-High club). Consequently, in the regions with high unemployment persistently surrounded by areas with similar performances, the adoption of well-designed "regional growth policies" are of key importance (also with respect to labor policies). This confirms the findings of the literature based on Okun's Law and on the key role of production and productivity dynamics (e.g., Marelli and Signorelli 2010b) in favoring improvement in the labor market. Moreover, an increase in the share of people employed in the manufacturing sector favors a reduction of unemployment in the whole of Russia, that is, for all the clubs/groups of regions. Hence, it seems that Russia has not yet reached the final stage of development, that is, when—according to the so-called "three sectors law"—the weight of industry starts to decline and the private services sector remains the only sector with growth in the share of both value added and employment. The policy implication is the adoption of industrial policies (e.g., with fiscal incentives for private investment in that sector) designed at national level and applied in all regions according to regional features and with the aim of reducing regional disparities in both development and employment performance.

To conclude, the various results obtained may be taken into account to design better economic policies at the national and regional level in order to reduce both overall unemployment and regional disparities. As said above, the policy impact on the unemployment rate in regions belonging to different clubs is probably quite different. In particular, regions belonging to the High-High club (mostly in the North Caucasus federal district and south of Siberia) differ significantly from other regions of Russia and require greater and better (national and regional) policy interventions in terms of growth and industrial policies and, to a lesser extent, with respect to the other regions' (active and passive) labor policies.

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Notes

1. In a pioneering research study, Overman and Puga (2002) clustered 150 NUTS-2 European regions according to several characteristics.
2. On the measurement of polarization, see Esteban and Ray (1994).
3. Few attempts have been made to combine the regional dimension and features with the national-level institutions (e.g., Perugini and Signorelli 2007).
4. For the Sakhalin region, these boundaries are measured by sea.
5. As suggested by one of the anonymous referees, this insignificant result could be partly explained by the well-known high percentage of people with higher education in Russia, jointly with a possible lower quality of tertiary education and, especially, a significant mismatch between the supply of graduates and the needs of companies.

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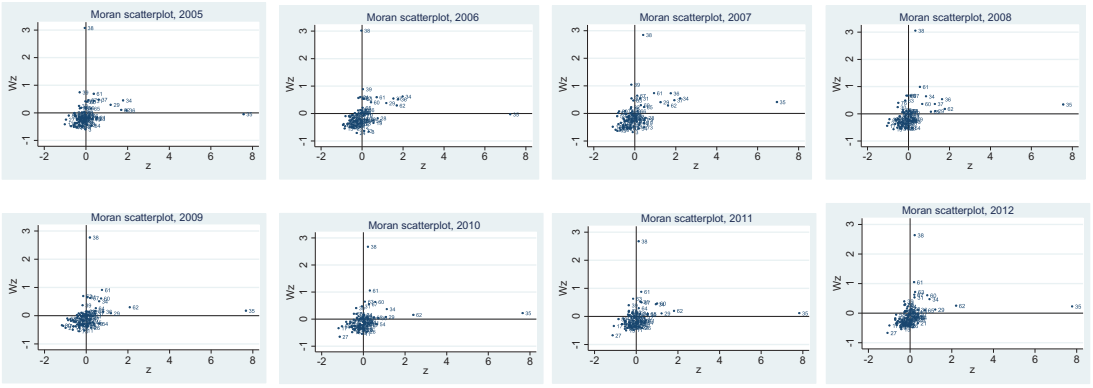
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Appendix A. The Moran's scatter plots for unemployment rates.



Appendix B. United subjects of the Russian Federation.

Data	Merging regions	Incorporated as
01.01.2007	Taymyr Autonomous Okrug	Krasnoyarsk Territory
	Evenk Autonomous Okrug	
	Krasnoyarsk territory	
01.07.2007	Kamchatka oblast	Kamchatka territory
	Koryak Autonomous Okrug	
01.01.2008	Ust-Orda Buryat Autonomous Okrug	Irkutsk region
	Irkutsk region	
01.03.2008	Chita region	Zabaykalsky Territory
	Aginsky Buryatsky Autonomous Okrug	
01.07.2012	Moscow	Moscow
	Moscow region	

Appendix C. Regional unemployment clubs in Russia in 2005–2012.

id	Region
Low-Low Club, $n_l = 46$	
1	Belgorod region
2	Bryansk region
3	Vladimir region
4	Voronezh region
5	Ivanovo region
6	Kaluga region
7	Kostroma region
8	Kursk region
9	Lipetsk region
10	Orel region
11	Ryazan region
12	Smolensk region
13	Tambov region
14	Tver region
15	Tula region
16	Yaroslavl region
17	Moscow
20	Arkhangelsk region
21	Nenets Autonomous Okrug
22	Vologda region
23	Leningrad region
24	Murmansk region
25	Novgorod region
26	Pskov region
27	Saint-Petersburg
40	Republic of Bashkortostan
42	Republic of Mordovia
43	Republic of Tatarstan
44	Republic of Udmurtia
46	Perm territory
47	Kirov region
48	Nizhny Novgorod region
49	Orenburg region
50	Penza region
51	Samara region
52	Saratov region
53	Ulyanovsk region
55	Sverdlovsk region
57	Khanty-Mansi Autonomous Area—Yugra
58	Yamal-Nenets autonomous region
59	Chelyabinsk region
70	Omsk region
74	Primorsky Territory
75	Khabarovsk Territory
77	Magadan region
80	Chukotka Autonomous Okrug
High-High Club, $n_l = 13$	
29	Republic of Kalmykia
31	Astrakhan region
34	Republic of Dagestan
35	Republic of Ingushetia
36	Republic of Kabardino—Balkaria
37	Republic of Karachaevo—Cherkessia
38	Republic of Northen Osetia—Alania
60	Republic of Altay
61	Republic of Buryatia
62	Republic of Tyva
64	Altay Territory
65	Zabaykalsky Territory
67	Irkutsk region
High-Low group, $n_{hl} = 21$	
18	Republic of Karelia
19	Republic of Komi
28	Republic of Adygea
30	Krasnodar Territory
32	Volgograd region
33	Rostov region
39	Stavropol Territory
41	Republic of Marii El
45	Republic of Chuvashia
54	Kurgan region
56	Tumen region
63	Republic of Khakassia
66	Krasnoyarsk Territory
68	Kemerovo region
69	Novosibirsk region
71	Tomsk region
72	Republic of Sakha (Yakutia)
73	Kamchatka territory
76	Amur region
78	Sakhalin region
79	Jewish autonomous area